ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 8

MATHEMATICS

PRACTICE EXAM 1 · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 8 Mathematics practice exam 1.

Written by Academic Central for practice, following the CAPS curriculum for Grade 8. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept other correct answers.

QUESTION 1

1 1.1 C ✓ 1.2 A ✓ 1.3 D ✓ 1.4 B ✓ 1.5 D ✓ 1.6 B ✓ 1.7 C ✓ 1.8 A ✓ 1.9 B ✓ 1.10 D ✓ [10]

QUESTION 2

2.1 $240 = 2^4 \times 3 \times 5$ ✓ $360 = 2^3 \times 3^2 \times 5$ ✓
$\text{HCF} = 2^3 \times 3 \times 5 = 120$ ✓ $\text{LCM} = 2^4 \times 3^2 \times 5 = 720$ ✓ (4)

2.2 $\text{SI} = 8\,000 \times 0{,}09 \times 4$ ✓ $= \text{R}2\,880$ ✓ $\text{Total} = \text{R}10\,880$ ✓ (3)

2.3.1 $10\% \times \text{R}6\,500 = \text{R}650$ ✓ (1)

2.3.2 $18 \times \text{R}390 = \text{R}7\,020$ ✓ $\text{R}650 + \text{R}7\,020 = \text{R}7\,670$ ✓ (2)

2.3.3 $\text{R}7\,670 - \text{R}6\,500 = \text{R}1\,170$ ✓ (1)

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QUESTION 3

3.1 (a) $40 + (-3)$ ✓ $= 37$ ✓
(b) $16 - 9$ ✓ $= 7$ ✓
(c) $-4 + 7$ ✓ $= 3$ ✓ (6)

3.2 (a) $4a^6$ ✓✓ (4 ✓, a^6 ✓) (b) $5x^4$ ✓ (c) $1 + 1 = 2$ ✓ (4)

[10]

QUESTION 4

4.1.1 $^7/_3 + {}^7/_4$ ✓ $= {}^{28}/_{12} + {}^{21}/_{12}$ ✓ $= {}^{49}/_{12} = 4\,{}^1/_{12}$ ✓ (3)

4.1.2 $^3/_5 \times {}^{10}/_9$ ✓ $= {}^{30}/_{45} = {}^2/_3$ ✓ (2)

4.1.3 $^4/_9 - {}^1/_9$ ✓ $= {}^3/_9 = {}^1/_3$ ✓ (2)

4.2 (a) 0,3 ✓ (b) $720 \div 9$ ✓ $= 80$ ✓ (3)

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QUESTION 5

5.1.1 Constant difference $= 3$ ✓ $\rightarrow \text{T}_n = 3n + 2$ ✓ (2)

5.1.2 $\text{T}_{25} = 3(25) + 2 = 77$ ✓ (1)

5.1.3 $3n + 2 = 152 \rightarrow 3n = 150$ ✓ $\rightarrow n = 50$; it is the 50th term ✓ (2)

5.2 (a) $-2(-2) + 1 = 5$ ✓ (b) $-2(2) + 1 = -3$ ✓ (c) $-2p + 1 = -9 \rightarrow p = 5$ ✓ (3)

5.3 (a) $5x^2$ ✓ $+ 2x$ ✓ (6)
(b) $-6x^2$ ✓ $+ 8x$ ✓
(c) $3a^2$ ✓ $- 2a$ ✓

5.4 (a) $3x = -15$ ✓ $\rightarrow x = -5$ ✓ (5)
(b) $x - 3 = 7$ (or $2x - 6 = 14$) ✓ $\rightarrow x = 10$ ✓
(c) $3^x = 3^4 \rightarrow x = 4$ ✓

5.5 $3x - 7 = x + 9 \rightarrow 2x = 16 \rightarrow x = 8$ ✓ (1)

[20]

QUESTION 6

6.1 (a) $\angle GHD = 65°$ ✓ (corresponding angles, AB ∥ CD) ✓ (6)
(b) $\angle BGH = 180° - 65° = 115°$ ✓ (adjacent angles on a straight line) ✓
(c) $\angle GHC = 180° - 65° = 115°$ ✓ (adjacent angles on a straight line; or alternate angles with $\angle BGH$, AB ∥ CD) ✓

6.2 $2x + 3x + 4x = 180°$ (sum of the angles of a triangle) ✓ $\rightarrow 9x = 180° \rightarrow x = 20°$ ✓ (3)
$\angle P = 40°, \angle Q = 60°, \angle R = 80°$ ✓

6.3 $\angle KMN = 48° + 67° = 115°$ ✓ (the exterior angle of a triangle equals the sum of the two interior opposite angles) ✓ (2)

6.4 (a) SSS ✓ (b) SAS ✓ (2)

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QUESTION 7

7.1 $h^2 = 5^2 - 1{,}4^2$ ✓ $= 25 - 1{,}96 = 23{,}04$ ✓ $\rightarrow h = 4{,}8$ m ✓ (3)

7.2 (a) $A = \pi r^2 = {}^{22}/_{7} \times 14 \times 14$ ✓ $= 616$ cm^2 ✓ (4)
(b) $C = 2\pi r = 2 \times {}^{22}/_{7} \times 14$ ✓ $= 88$ cm ✓

7.3 (a) V = area of cross-section × length $= ½ \times 3 \times 4 \times 10$ ✓ $= 60$ cm^3 ✓ (5)
(b) $SA = 2(½ \times 3 \times 4) + (3 + 4 + 5) \times 10$ ✓ $= 12 + 120$ ✓ $= 132$ cm^2 ✓

[12]

QUESTION 8

8.1 8.1.1 (3 ; 2) ✓ 8.1.2 (−3 ; −2) ✓ 8.1.3 (−1 ; 1) ✓✓ (one mark per coordinate) 8.1.4 (−2 ; 3) ✓ 8.1.5 (6 ; −4) ✓ (6)

[6]

QUESTION 9

9.1.1 $600 \div 10$ ✓ $= 60$ ✓ (2)

9.1.2 Ordered: 39 ; 45 ; 55 ; 58 ; 62 ; 62 ; 62 ; 66 ; 71 ; 80 ✓ → median $= (62 + 62) \div 2 = 62$ ✓ (2)

9.1.3 62 ✓ (1)

9.1.4 $80 - 39 = 41$ ✓ (1)

9.2 (a) $^{12}/_{60} = {}^{1}/_{5}$ ✓ (b) $^{1}/_{6}$ ✓ (2)

[8]

TOTAL: 100

END OF MARKING GUIDELINE